Curvature and Torsion in the Continuum Theory of Lattice Defects

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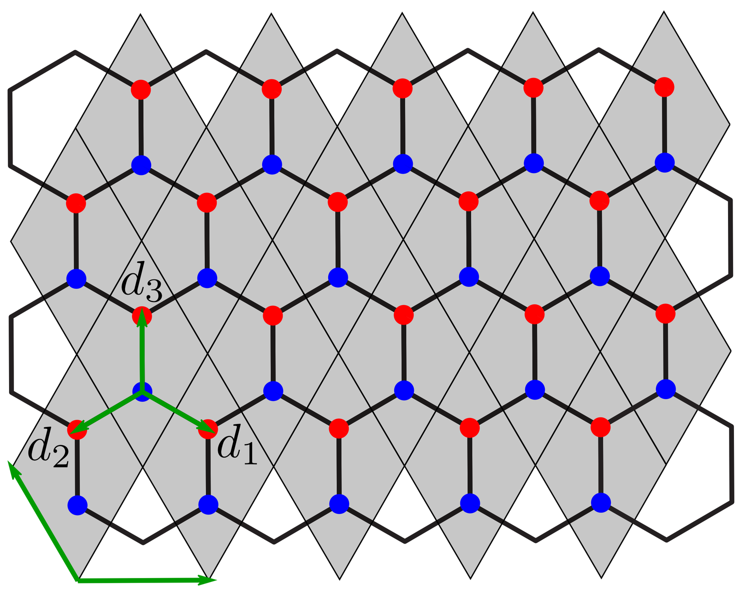
Continuum Theory - Graphene[1]



Tight-binding Hamiltonian

$$H = \sum_{\vec{r}} \sum_{j=1}^{3} \left(a^\dagger_{\vec{r}} b_{\vec{r}+\vec{d}_j} + b^\dagger_{\vec{r}+\vec{d}_j} a_{\vec{r}} \right)$$

Fourier transform

$$i\partial_t \vec{\psi}(\vec{k}) = \boldsymbol{h}(\vec{k})\vec{\psi}(\vec{k})$$

$$\boldsymbol{h}(\vec{k}) = \begin{pmatrix} 0 & f(\vec{k}) \\ f^*(\vec{k}) & 0 \end{pmatrix}, \quad f(\vec{k}) = \sum_{j=1}^{3} e^{i\vec{k}\cdot\vec{d}_j}$$

Linear dispersion around Dirac points

$$\vec{k} = \vec{K}^{(s)} + \delta\vec{k} \quad \Longrightarrow \quad \boldsymbol{h}(\vec{k}) \approx -\frac{3}{2}\delta\vec{k}\cdot\vec{\boldsymbol{\sigma}}$$

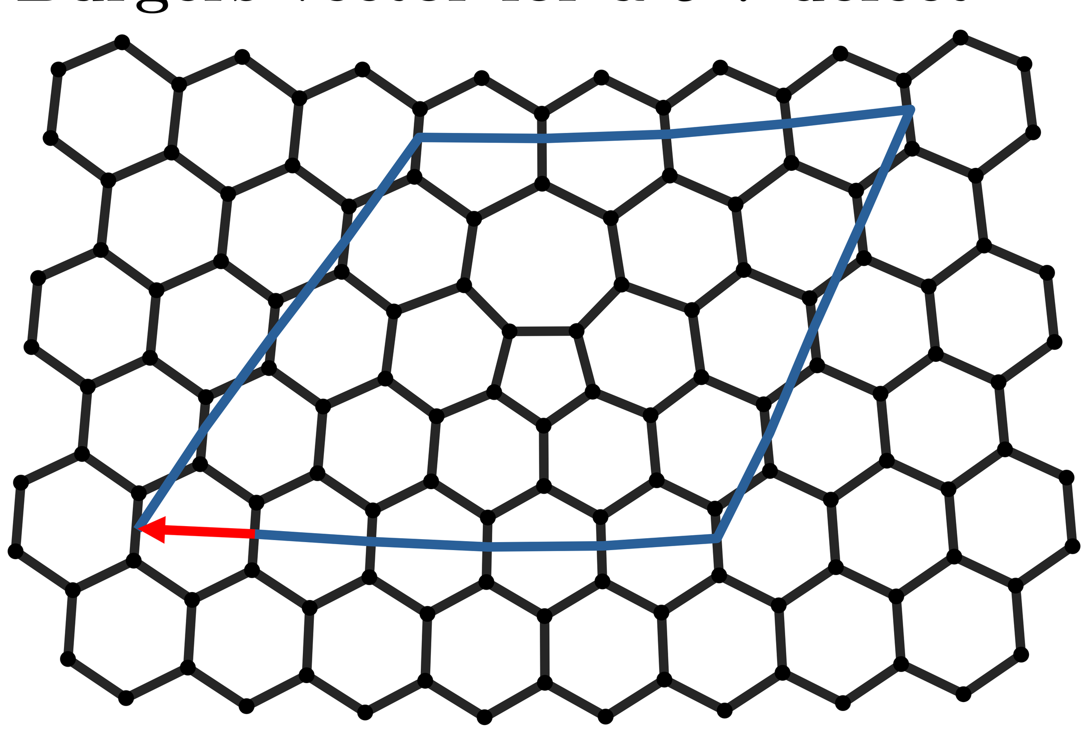
Continuum limit

$$i\partial_t \vec{\phi}(\vec{r}) = i\, c\, \boldsymbol{\sigma}^i \partial_i \vec{\phi}(\vec{r}), \qquad \rho = \vec{\phi}^\dagger \vec{\phi}, \qquad \vec{j} = -c\vec{\phi}^\dagger \vec{\boldsymbol{\sigma}} \vec{\phi}$$

$$\Updownarrow$$

$$i\gamma^\mu \partial_\mu \phi = 0, \quad \gamma^\mu = \sigma^\mu \sigma^3, \quad \mu \in \{0,1,2\}$$

Burgers vector for a 5-7 defect[2]



▶ **Lattice dislocation** with Burgers vector:

$$\vec{b} = -\sqrt{3}\,\vec{u}_x$$

Torsion[3]

▶ Standard GR: torsion vanishes
▶ Torsion = **antisymmetric** part of the connection

$$T^\lambda{}_{\mu\nu} = \Gamma^\lambda{}_{\mu\nu} - \Gamma^\lambda{}_{\nu\mu}$$

▶ Decomposition of any general affine connection into symmetric part, contorsion & disformation:

$$\Gamma^\alpha{}_{\mu\nu} = \{^\alpha{}_{\mu\nu}\} + K^\alpha{}_{\mu\nu} + L^\alpha{}_{\mu\nu}$$

▶ **Metric-compatibility**:

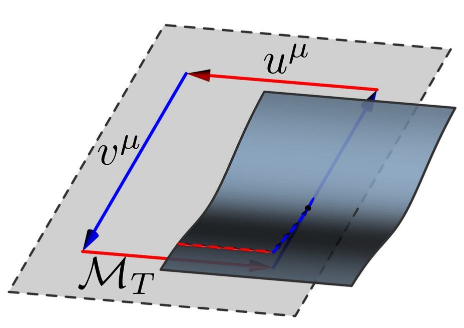
$$\nabla_\lambda g_{\mu\nu} = 0 \Leftrightarrow L^\alpha{}_{\mu\nu} = 0$$

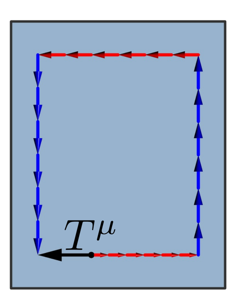
▶ **Contorsion tensor** from torsion:

$$K^\alpha{}_{\mu\nu} = \frac{1}{2}\left(T^\alpha{}_{\mu\nu} - T_{\mu\nu}{}^\alpha + T_\nu{}^\alpha{}_\mu\right)$$

▶ **Visual** idea: displacement vector $T^\mu(u,v)$ when rolling tangent plane along parallelogram:

$$T^\mu(u,v) = u^\lambda v^\nu T^\mu{}_{\nu\lambda}$$



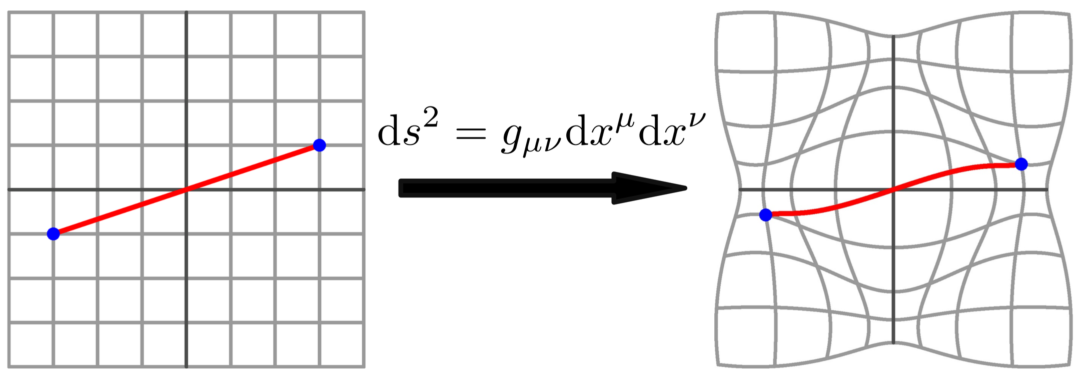


Riemannian Geometry[3]

▶ Riemannian **manifold** $\mathcal{M}$ with metric $g_{\mu\nu}$ and connection $\Gamma^\lambda{}_{\mu\nu}$

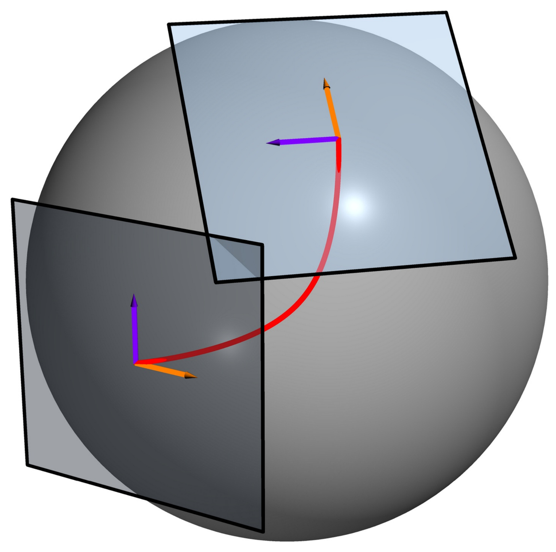
Metric

▶ Defines distances & angles on manifold



Connection

▶ Connectes nearby **tangent spaces**
▶ Allows definition of **tensorial** derivative
▶ Unique torsion-free, metric-compatible connection:
Levi-Civita connection



$$\{^\lambda_{\mu\nu}\} = \frac{1}{2} g^{\lambda\rho} \left(\partial_\nu g_{\rho\mu} + \partial_\mu g_{\rho\nu} - \partial_\rho g_{\mu\nu}\right)$$

Covariant derivative

▶ Takes into account the **structure** of the manifold
▶ Transforms like a **tensor**

$$\nabla_\mu v^\nu = \partial_\mu v^\nu + \Gamma^\nu{}_{\mu\lambda} v^\lambda$$

Autoparallels

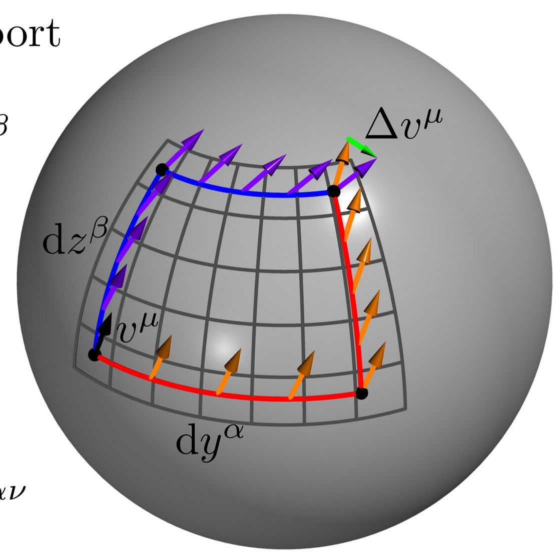
▶ Tangent vector is parallel-transported

$$u^\mu \nabla_\mu u^\alpha = 0$$

Curvature

▶ how does parallel transport depend on the path?
▶ $\Delta v^\mu = R_{\alpha\beta}{}^\mu{}_\nu\, v^\nu\, dy^\alpha\, dz^\beta$
▶ Fully determined by the connection:

$$R_{\alpha\beta}{}^\mu{}_\nu = + \partial_\alpha \Gamma^\mu{}_{\beta\nu} - \partial_\beta \Gamma^\mu{}_{\alpha\nu} + \Gamma^\mu{}_{\alpha\lambda}\Gamma^\lambda{}_{\beta\nu} - \Gamma^\mu{}_{\beta\lambda}\Gamma^\lambda{}_{\alpha\nu}$$



Metric ↔ torsion mapping in 2D

▶ Physical mapping between:
 ▶ **Classical** Manifold $\mathcal{M}_g(g_{ij}, \{^k_{ij}\})$
 ▶ **Purely torsionful** Manifold $\mathcal{M}_T(\delta_{ij}, K^k{}_{ij}\})$
▶ Coordinate choice: $g_{\mu\nu} = \delta_{\mu\nu}\, e^{2\phi(x,y)}$.
▶ Antisymmetry: $T^k{}_{ij} = b^k \varepsilon_{ij}$
▶ Demand equal **curvature** of both manifolds:

$$R\left[\mathcal{M}_g\right] = R\left[\mathcal{M}_T\right],$$

▶ leads to solution for torsion vector b^i as function of metric ϕ and arbitrary scalar field χ:

$$b^i = \varepsilon_{ik}\, \partial_k \phi + \partial_i \chi$$

▶ for $\chi = 0$: **autoparallels** of both manifolds equivalent up to reparameterization
▶ χ corresponds to local **roatations** of the frame field

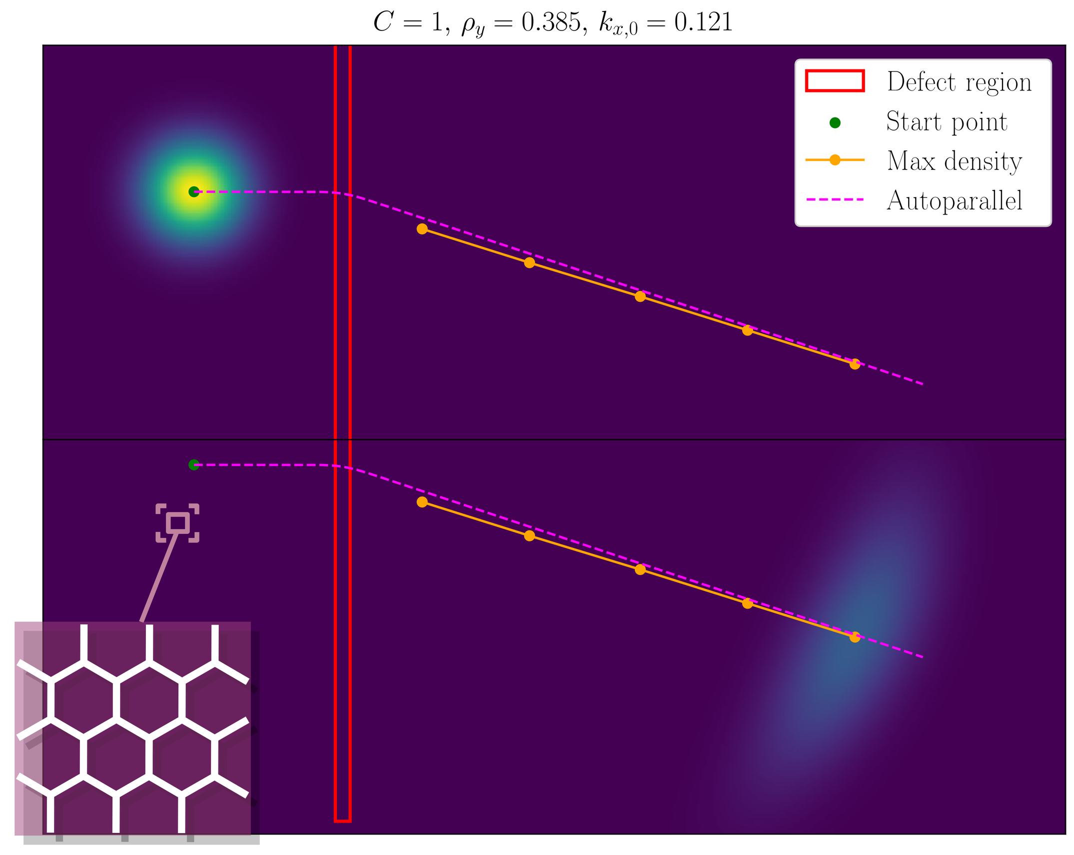
Conclusion

▶ Autoparallels and curvature can be mapped
▶ Different distance measure
▶ Other tangent spaces and tensors

Tight-binding numerics

▶ **vector representation** of wavefunction at discrete lattice sites
▶ Hamiltonian is (sparse) **Matrix** acting on this vector to generate time evolution:

$$i\partial_t \vec{\psi} = H\vec{\psi} \quad \Rightarrow \quad \vec{\psi}(t) = e^{-iHt}\vec{\psi}_0$$



Torsion model

▶ $\mathcal{M}_T$ with $b^1(x)$ and $b^2 \equiv 0$
▶ Autoparallel equation:

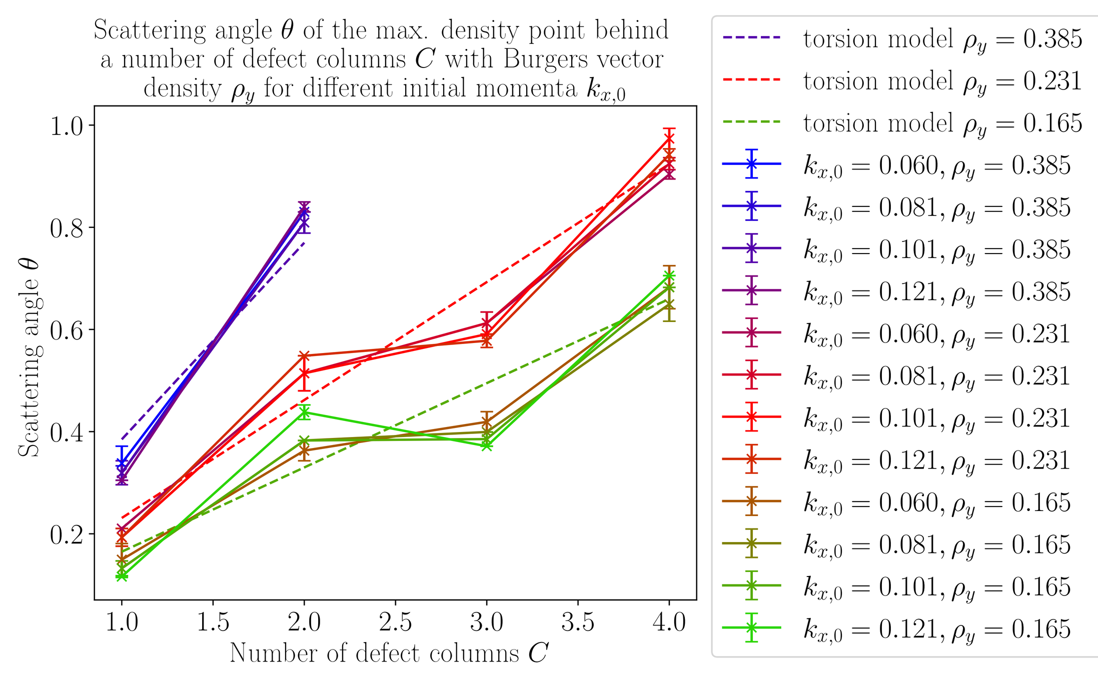
$$\ddot{x} = -\dot{x}\dot{y}b^1, \quad \ddot{y} = \dot{x}^2 b^1$$

▶ Allows calculating the **scattering angle**:

$$\theta = \int b^1(x)\, dx$$

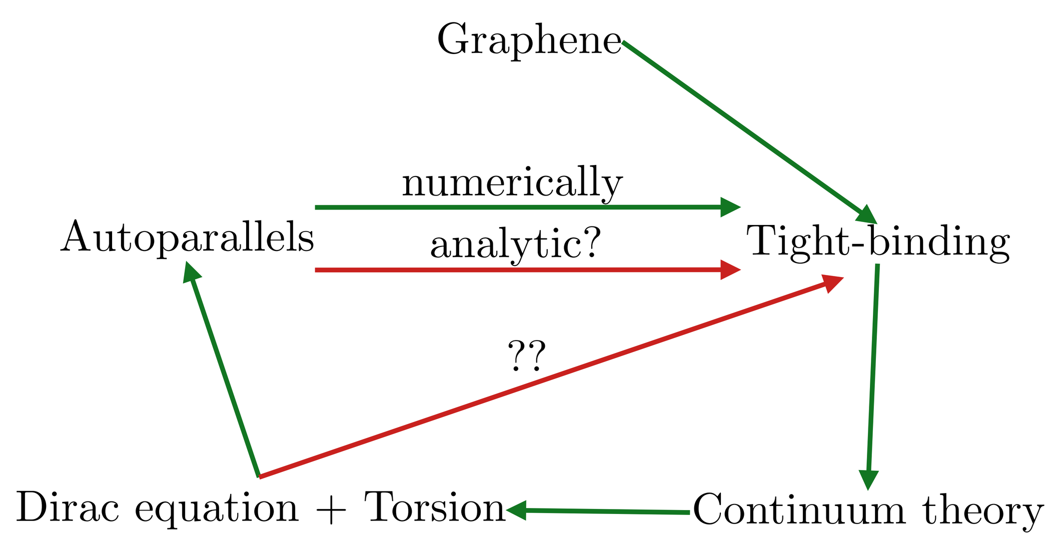
▶ Interpreting torsion as Burgers vector density:

$$\rho_y C = \frac{C\cdot(\vec{b})_x}{\Delta y} \approx \frac{\iint b^1 dx dy}{\Delta y} = \int b^1 dx = \theta$$



▶ Torsion model predicts linear relation $\theta = \rho_y C$
▶ Tight-binding simulations match prediction from torsion model with autoparallels

Conclusion & Outlook[4]



▶ Can the torsion interpretation of defects via autoparallels be shown analytically?
▶ Is it possible to describe the tight-binding results with the Dirac equation coupled to torsion?
▶ Does the phyisical mapping between torsion and metric apply to the description via torsion?

References

[1] Stegmann T, Szpak N. Current flow paths in deformed graphene. *New J. Phys.* **18**, 053016 (2016).
[2] Bhatt MD, Kim H, Kim G. Various defects in graphene: a review. *RSC Adv.* **12**, 21520 (2022).
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[4] de Juan F, Cortijo A, Vozmediano MAH. Dislocations and torsion in graphene. *Nucl. Phys. B* **828**, 625 (2010).